Harsh Raj

8294856748 | harshishu2005@gmail.com | linkedin.com/in/harshraj0005 | github.com/harshraj005

PROFESSIONAL SUMMARY

Backend engineer focused on Core Java, multi-threading, and data security. Combines a rigorous academic background with practical success in national-level hackathons and open-source development. Excels at tackling complex algorithmic challenges—whether designing concurrent data structures, optimizing cryptographic file I/O, or architecting secure backend components from first principles.

EDUCATION

Shobhit Institute of Engineering & Technology — Meerut, Uttar Pradesh
Bachelor of Technology in Computer Science and Engineering | CGPA: 8.37/10.0 — Aug 2024 – Present

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, Software Engineering.

R.K.D.F University — Ranchi, Jharkhand
Diploma in Computer Science and Engineering | CGPA: 8.63/10.0 — Oct 2021 – June 2024

PROJECTS

Thread-Safe LRU Cache | *Core Java, Multithreading, Concurrency* — View on GitHub

- Built a generic, thread-safe in-memory LRU cache in core Java using a HashMap + doubly-linked list to achieve O(1) get and put operations with configurable capacity and TTL expiry.
- Implemented concurrent access using ReentrantReadWriteLock, allowing parallel reads and exclusive writes; used AtomicLong for lock-free hit/miss/eviction stats tracking.
- Designed a fluent Builder API and a background daemon sweeper via ScheduledExecutorService to lazily evict TTL-expired entries without blocking the hot path.
- Wrote a zero-dependency test suite with 10 tests covering eviction order, TTL boundaries, and a concurrent stress test using CountDownLatch across 8 simultaneous threads.

Hybrid AES-RSA Encryption Engine | *Core Java, Cryptography, File I/O* — View on GitHub

- Developed a robust cryptographic engine in Core Java capable of securing large files by combining the speed of AES symmetric encryption with the secure key-exchange mechanism of RSA.
- Leveraged standard `java.security` and `javax.crypto` libraries to implement secure padding schemes, ensuring data integrity and resolving common vulnerabilities like predictable initialization vectors (IVs).
- Optimized the file parsing and encryption loop using byte-array buffering and stream handling, preventing memory overflow and significantly reducing processing time for large data payloads.

Smart Meeting Room Scheduler | *Core Java, OOP, Collections* — View on GitHub

- Engineered a terminal-based application in Core Java to manage meeting room availability, handle user reservations, and resolve scheduling conflicts in real-time.
- Leveraged the Java Collections Framework to efficiently store and retrieve scheduling data and room properties in memory.
- Implemented robust exception handling and string parsing logic to validate user inputs, preventing overlapping schedules and ensuring data integrity.

OPEN SOURCE CONTRIBUTIONS

Text Encryption & Decryption Tool | *Social Summer of Code* — 2025

- Resolved 2 critical RSA encryption bugs (duplicate keys, incorrect padding) and enhanced functionality by integrating Caesar cipher support with UI improvements using JavaScript, HTML, and CSS.
- Achieved 'Advanced' contributor status by maintaining a 100% issue resolution rate and reducing code review turnaround time by ∼30% through efficient Agile workflows.

TECHNICAL & SOFT SKILLS

Languages: Core Java, C, C++, Python, SQL, Kotlin, JavaScript
Backend & Core Concepts: Multithreading, Concurrency (Locks, Atomics), Java Collections Framework, OOP, Memory Management
Security & Cryptography: AES/RSA Symmetric & Asymmetric Encryption, Secure Key Exchange, Data Security
Developer Tools: Linux, Postman, VS Code, Git, GitHub
Soft Skills: Analytical Problem Solving, Algorithmic Thinking, Effective Communication, Agile Adaptability, Time Management

ACHIEVEMENTS

- **Research Publication:** Co-Authored "Integrating Vedic Mathematics in Post-Quantum Cryptography." Abstract published; full research paper communicated for publication.
- **Winner, Internal Smart India Hackathon 2025:** Selected from University for national-level innovation showcase.
- **Runner-Up, Reverse Coding Competition 2024:** Secured top 5% placement by decoding complex logic patterns.
- **Contributor, Social Summer of Code (SSoC) 2025:** Recognized for advanced-level open-source contributions.
- **Gold Medalist, Diploma in CSE:** Awarded for securing the top rank in every semester.